Name:

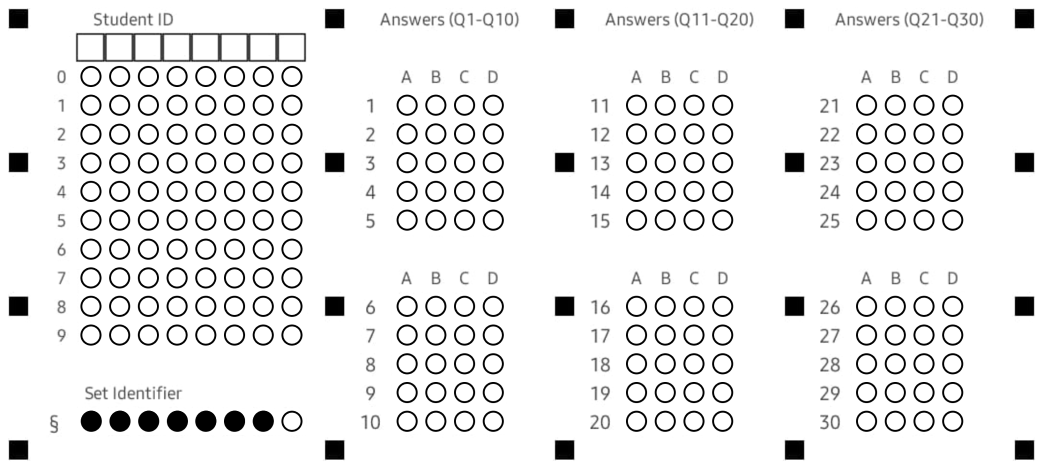
Brac University
Semester: Spring 2026
Course Code: CSE250
Circuits and Electronics
Section: 01
Faculty: SDS

Set
A

Assessment: *Main Template*
Duration: 30 minutes
Date: May 4, 2026
Full Marks: 10

- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ All **4 question(s)** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Write your answers inside the indicated boxes (where applicable). If you run out of room, continue on the back page.
- ✓ Symbols have their usual meanings.

DO NOT WRITE ANYTHING ELSE INSIDE THIS BORDER



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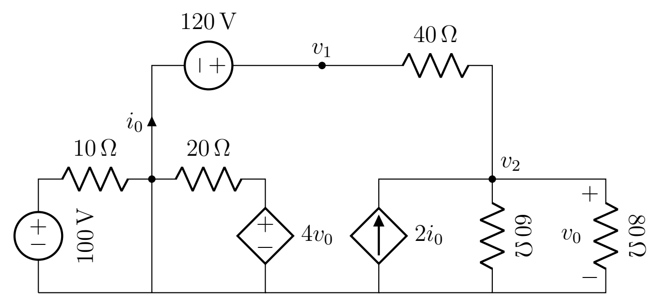
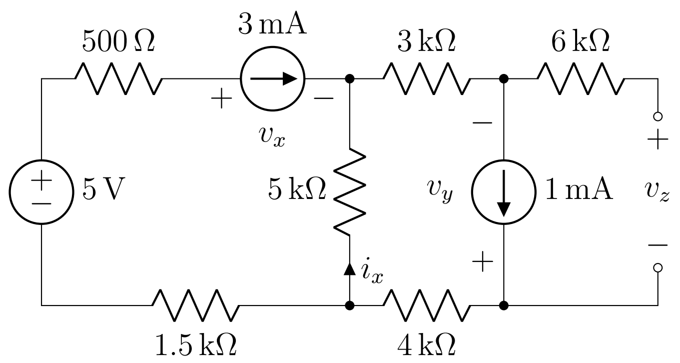
◇ Question 1 of 4 *[CO1] [1 mark]*



How many nodes are in this circuit?

(a) 5 **(b)** 3 **(c)** 4 **(d)** 7 **(e)** 6

◇ Question 2 of 4 *[CO3] [2 marks]*

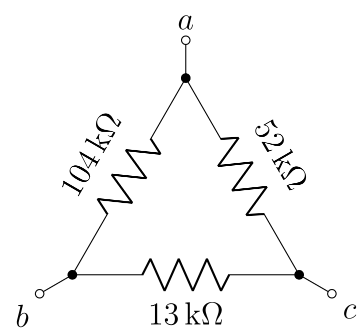


(a) c

(b) b

(c) a

◇ Question 3 of 4 *[CO3] [5 marks]*



Which of the following statements is true?

(a) $R_{ab} = 40\,k\Omega, R_{bc} = 12\,k\Omega, R_{ca} = 36\,k\Omega$

(b) $R_{ab} = R_{bc} = R_{ca} = 9.455\,k\Omega$

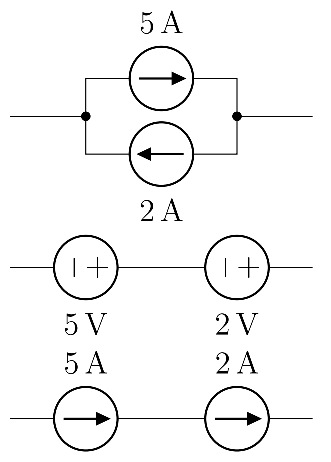
(c) $R_{ab} = 40\,\Omega, R_{bc} = 12\,\Omega, R_{ca} = 36\,\Omega$

(d) $R_{ab} = 104\,k\Omega, R_{bc} = 13\,k\Omega, R_{ca} = 52\,k\Omega$

(e) None of the above

Which of the following circuits is/are impossible (violates Kirchhoff's laws)?

♢ **Question 4 of 4** ***[CO1] [2 marks]***



Circuit 1

Circuit 2

Circuit 3

(a) Circuit 2 **(b)** Circuit 1 & 3 **(c)** Circuit 1 **(d)** Circuit 3 **(e)** None of the aboveDo you have any suggestion that may improve this course?

Your Suggestions

.... ↓ *Your answers/roughs start here* ↓ ....

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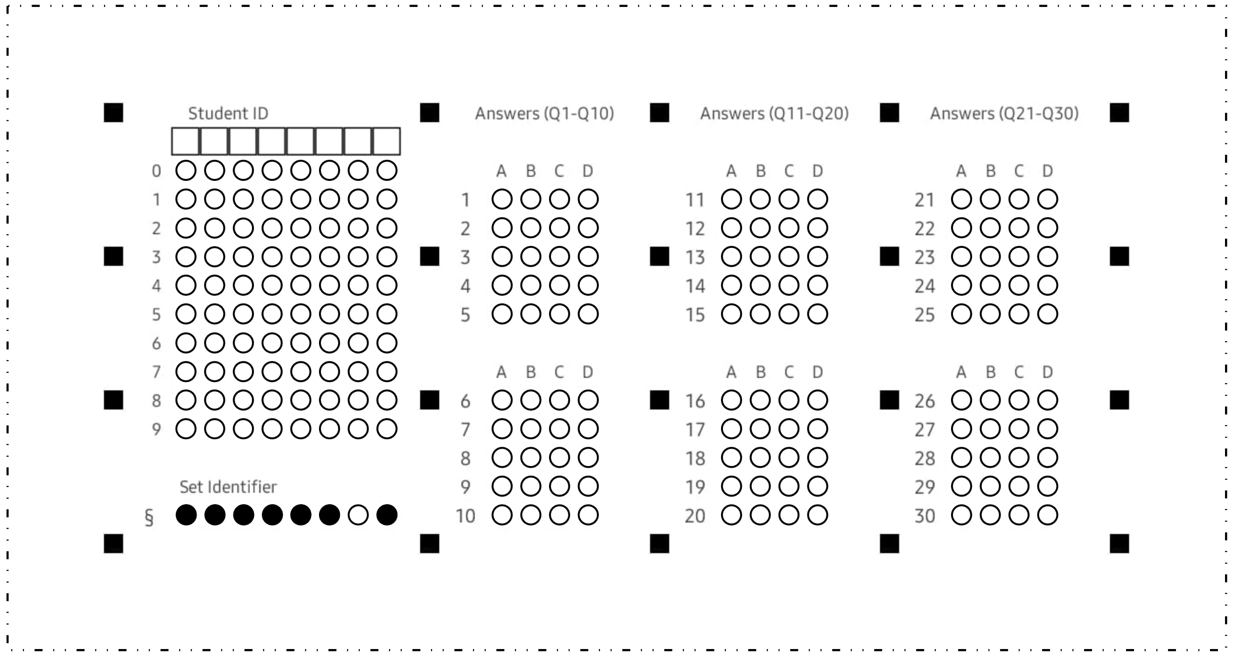

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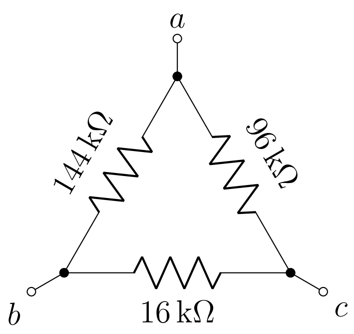
◇ **Question 1 of 4** ***[CO3] [5 marks]***



Which of the following statements is true?

(a) $R_{ab} = R_{bc} = R_{ca} = 12.522\,k\Omega$

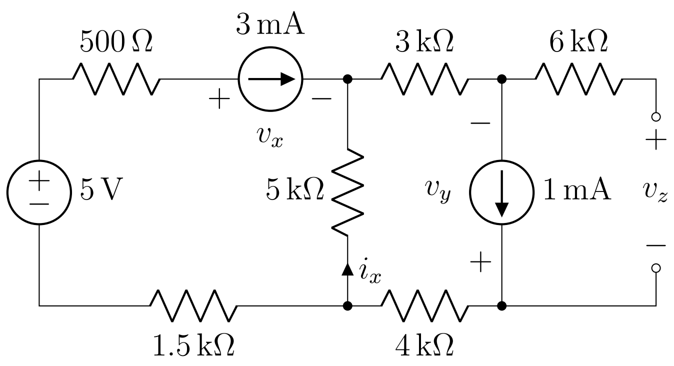
(b) $R_{ab} = 63\,k\Omega, R_{bc} = 15\,k\Omega, R_{ca} = 60\,k\Omega$

(c) $R_{ab} = 63\,\Omega, R_{bc} = 15\,\Omega, R_{ca} = 60\,\Omega$

(d) $R_{ab} = 144\,k\Omega, R_{bc} = 16\,k\Omega, R_{ca} = 96\,k\Omega$

(e) None of the above

◇ **Question 2 of 4** ***[CO3] [2 marks]***

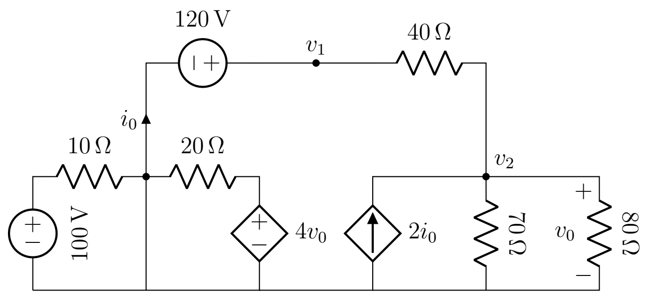


(a) c

(b) b

(c) a

◇ **Question 3 of 4** ***[CO1] [1 mark]***

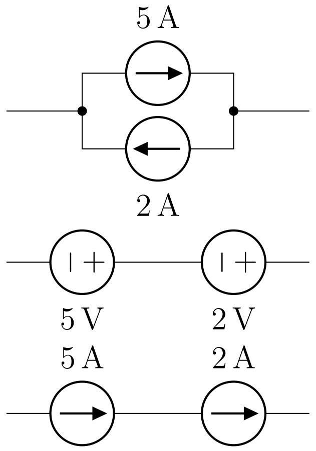


How many nodes are in this circuit?

(a) 6 **(b)** 4 **(c)** 3 **(d)** 5 **(e)** 7

◇ **Question 4 of 4** ***[CO1] [2 marks]***

Which of the following circuits is/are impossible (violates Kirchhoff's laws)?



Circuit 1

Circuit 2

Circuit 3

(a) Circuit 3 **(b)** Circuit 1 **(c)** Circuit 1 & 3 **(d)** Circuit 2 **(e)** None of the aboveDo you have any suggestion that may improve this course?

Your Suggestions

.... ↓ *Your answers/roughs start here* ↓ ....

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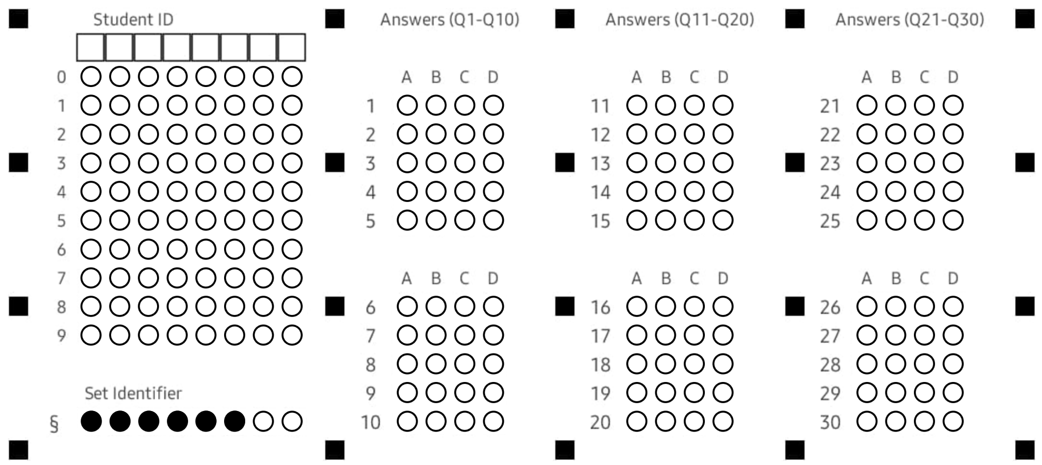

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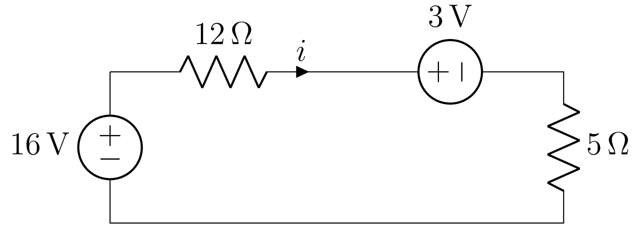
◇ Question 1 of 4 *[CO3] [2 marks]*



What's the relation between v_1, v_2 and i?

(a) $-16 + 12i - 3 + 5i = 0$

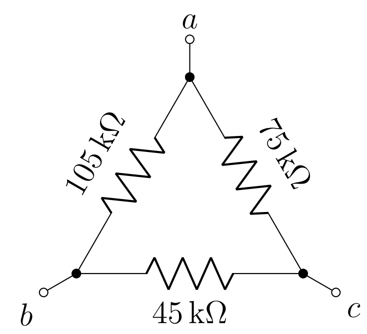
(b) $-16 - 12i - 3 + 5i = 0$

(c) $-16 - 12i + 3 + 5i = 0$

(d) $-16 + 12i + 3 + 5i = 0$

(e) None of the above

◇ Question 2 of 4 *[CO3] [5 marks]*



Which of the following statements is true?

(a) $R_{ab} = R_{bc} = R_{ca} = 22.183\,\mathrm{k\Omega}$

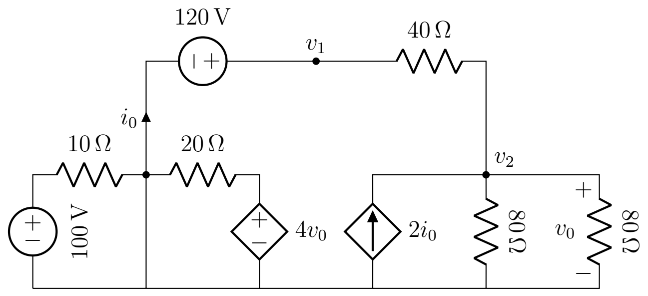
(b) $R_{ab} = 56\,\Omega, R_{bc} = 36\,\Omega, R_{ca} = 50\,\Omega$

(c) $R_{ab} = 105\,\mathrm{k\Omega}, R_{bc} = 45\,\mathrm{k\Omega}, R_{ca} = 75\,\mathrm{k\Omega}$

(d) $R_{ab} = 56\,\mathrm{k\Omega}, R_{bc} = 36\,\mathrm{k\Omega}, R_{ca} = 50\,\mathrm{k\Omega}$

(e) None of the above

◇ **Question 3 of 4** ***[CO1] [1 mark]***

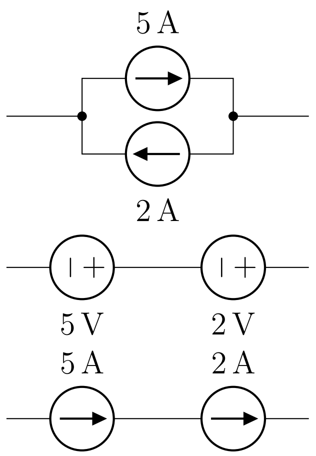


How many nodes are in this circuit?

(a) 7 **(b)** 5 **(c)** 3 **(d)** 4 **(e)** 6

◇ **Question 4 of 4** ***[CO1] [2 marks]***

Which of the following circuits is/are impossible (violates Kirchhoff's laws)?



Circuit 1

Circuit 2

Circuit 3

(a) Circuit 1 **(b)** Circuit 1 & 3 **(c)** Circuit 2 **(d)** Circuit 3 **(e)** None of the aboveDo you have any suggestion that may improve this course?

Your Suggestions

.... ↓ *Your answers/roughs start here* ↓ ....

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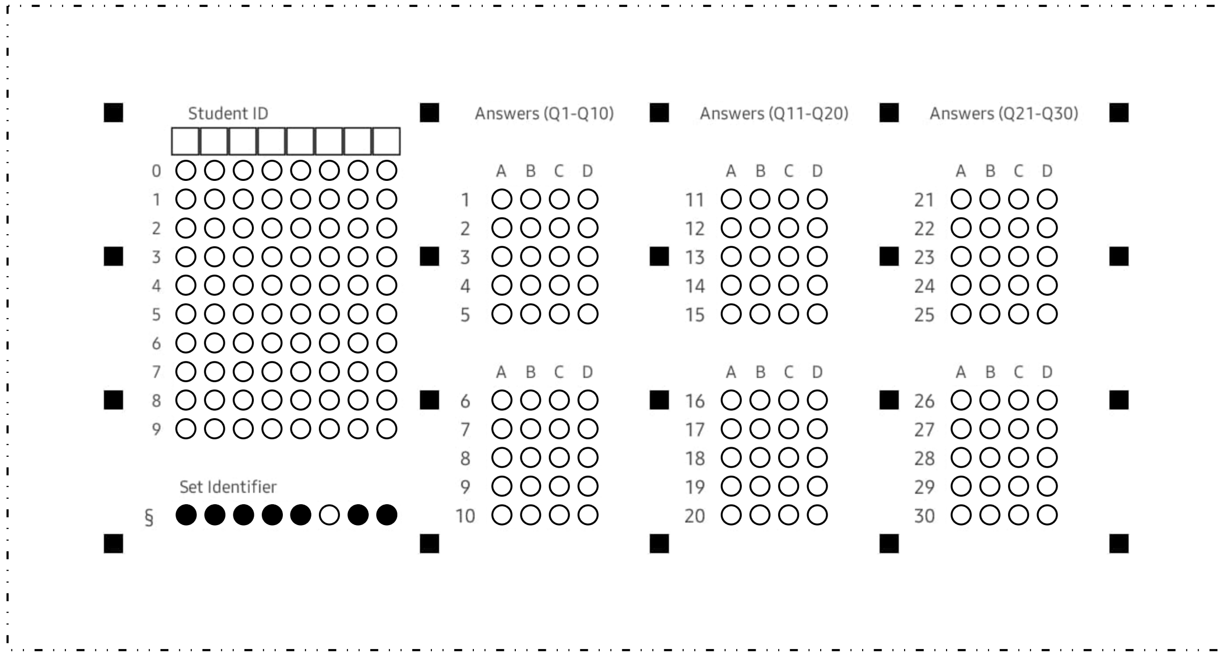

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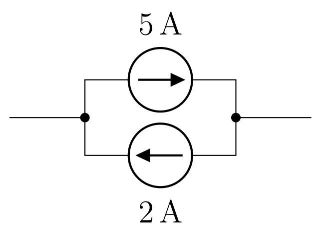
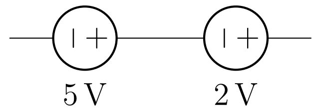
◇ **Question 1 of 4** ***[CO1] [2 marks]***

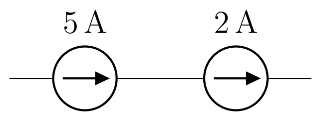
Which of the following circuits is/are impossible (violates Kirchhoff's laws)?



Circuit 1



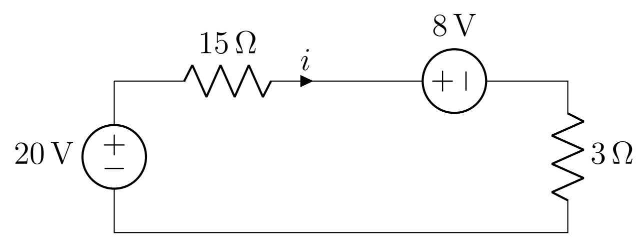
Circuit 2



Circuit 3

(a) Circuit 1 & 3 **(b)** Circuit 2 **(c)** Circuit 1
(d) Circuit 3 **(e)** None of the above

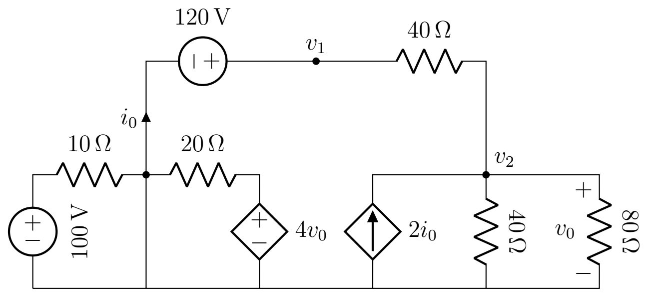
◇ **Question 2 of 4** ***[CO3] [2 marks]***



What's the relation between v_1, v_2 and i?

- **(a)** $-20 + 15i + 8 + 3i = 0$
- **(b)** $-20 - 15i - 8 + 3i = 0$
- **(c)** $-20 + 15i - 8 + 3i = 0$
- **(d)** $-20 - 15i + 8 + 3i = 0$
- **(e)** None of the above

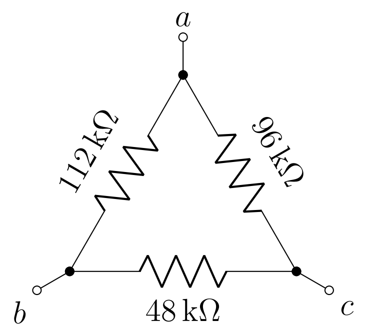
◇ **Question 3 of 4** ***[CO1] [1 mark]***



How many nodes are in this circuit?

(a) 6 **(b)** 5 **(c)** 3 **(d)** 7 **(e)** 4

◇ **Question 4 of 4** ***[CO3] [5 marks]***



Which of the following statements is true?

- **(a)** $\mathrm{R}_{ab} = \mathrm{R}_{bc} = \mathrm{R}_{ca} = 24.889\,\mathrm{k}\Omega$
- **(b)** $\mathrm{R}_{ab} = 63\,\mathrm{k}\Omega, \mathrm{R}_{bc} = 39\,\mathrm{k}\Omega, \mathrm{R}_{ca} = 60\,\mathrm{k}\Omega$
- **(c)** $\mathrm{R}_{ab} = 63\,\Omega, \mathrm{R}_{bc} = 39\,\Omega, \mathrm{R}_{ca} = 60\,\Omega$
- **(d)** $\mathrm{R}_{ab} = 112\,\mathrm{k}\Omega, \mathrm{R}_{bc} = 48\,\mathrm{k}\Omega, \mathrm{R}_{ca} = 96\,\mathrm{k}\Omega$
- **(e)** None of the above

Do you have any suggestion that may improve this course?

Your Suggestions

.... ↓ *Your answers/roughs start here* ↓ ....